

Heteroskedasticity

Intermediate Econometrics

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Wooldridge, Chapter 8

Outline

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Motivation and Definition (§8-0)

What Is Heteroskedasticity?

Recall **MLR.5 (Homoskedasticity)**:

$$\text{Var}(u \mid x_1, \dots, x_k) = \sigma^2 \quad (\text{constant for all observations})$$

Heteroskedasticity

MLR.5 is **violated** when the error variance depends on the regressors:

$$\text{Var}(u_i \mid x_{i1}, \dots, x_{ik}) = \sigma_i^2$$

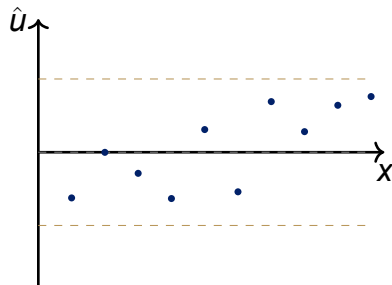
where σ_i^2 **varies** across observations $i = 1, \dots, n$.

Why does this happen?

- ▶ Richer households have *more variable* consumption (income effect on spread)

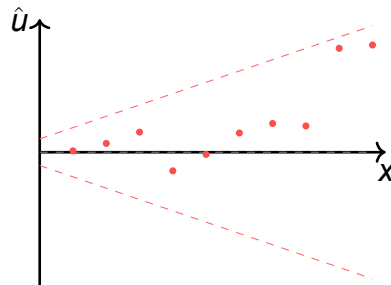
Visualising Heteroskedasticity

Homoskedastic errors



Constant band

Heteroskedastic errors



Fan-shaped spread

Residual plots ($\hat{u}_i$ vs. $\hat{y}_i$ or $\hat{u}_i^2$ vs. x_i) are the first diagnostic. A funnel or fan shape signals heteroskedasticity.

Consequences for OLS (§8-1)

Consequences: What Changes, What Stays the Same

NOT affected:

- ▶ **Unbiasedness** of $\hat{\beta}_j$ (only MLR.1–4 needed)
- ▶ **Consistency** of $\hat{\beta}_j$
- ▶ Interpretation of R^2 and $\bar{R}^2$

⇒ OLS point estimates are *still useful*.

IS affected:

- ▶ OLS **standard errors** are wrong
- ▶ ***t*-tests** and ***F*-tests** are invalid
- ▶ OLS is **no longer BLUE** (Gauss-Markov fails)
- ▶ **Efficiency** is lost

⇒ Inference based on usual SEs is *unreliable*.

Why OLS Standard Errors Are Wrong: The Proof

True variance of $\hat{\beta}_1$ in simple regression (derived from $\hat{\beta}_1 = \beta_1 + \sum_i w_i u_i$):

$$\text{Var}(\hat{\beta}_1) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma_i^2}{\text{SST}_x^2} \quad (\text{eq. 8.2})$$

where $\text{SST}_x = \sum_i (x_i - \bar{x})^2$.

Under homoskedasticity ($\sigma_i^2 = \sigma^2$ for all i), this simplifies to:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2 \sum_i (x_i - \bar{x})^2}{\text{SST}_x^2} = \frac{\sigma^2}{\text{SST}_x}$$

OLS uses $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$ to estimate σ^2 . If $\sigma_i^2 \neq \sigma^2$, then $\hat{\sigma}^2$ estimates a **weighted average** of σ_i^2 , not the right quantity for eq. 8.2.

Direction of bias: *unpredictable* without knowing how σ_i^2 relates to leverage $(x_i - \bar{x})^2$.

Gauss-Markov Fails: OLS Is No Longer BLUE

Recall the Gauss-Markov Theorem requires MLR.1–5. Under heteroskedasticity:

There exist **other linear unbiased estimators** with *smaller variance* than OLS.

Specifically, **Weighted Least Squares** (WLS) with the correct weights achieves lower variance than OLS.

Intuition:

- ▶ Observations with *small* σ_i^2 are very informative — OLS under-uses them.
- ▶ Observations with *large* σ_i^2 are noisy — OLS over-uses them.
- ▶ WLS corrects this by weighting inversely proportional to variance.

Even though OLS estimates remain unbiased, an analyst using OLS un-

Heteroskedasticity-Robust Inference (§8-2)

The White Robust Variance Estimator

Key insight (White, 1980): Replace the unknown σ_i^2 in eq. 8.2 with $\hat{u}_i^2$ (the squared OLS residual for observation i).

Heteroskedasticity-Robust Variance Estimator (eq. 8.4)

$$\widehat{\text{Var}}_{\text{rob}}(\hat{\beta}_j) = \frac{\sum_{i=1}^n \hat{r}_{ij}^2 \hat{u}_i^2}{\text{SSR}_j^2}$$

- ▶ $\hat{r}_{ij}$ = residuals from regressing x_j on *all other regressors*
- ▶ $\text{SSR}_j = \sum_i \hat{r}_{ij}^2$
- ▶ $\hat{u}_i$ = OLS residuals from the main regression

▶ Also called **White SE**, **Hetero-White SE**, **HQ standard errors**

Why the Robust Estimator Is Consistent: Sketch

We need: A consistent estimator of $\sum_i \hat{r}_{ij}^2 \sigma_i^2 / \text{SSR}_j^2$.

Step 1. Note that $\text{plim } n^{-1} \text{SSR}_j = \text{Var}(r_j)$ (a constant), so SSR_j^2 grows at rate n^2 .

Step 2. For each i : $\hat{u}_i = u_i + (\beta - \hat{\beta})'x_i$. Under MLR.1–4 and consistency of $\hat{\beta}$:

$$\hat{u}_i^2 \stackrel{p}{\approx} u_i^2 \quad \text{as } n \rightarrow \infty$$

Step 3. Therefore:

$$\frac{1}{n} \sum_i \hat{r}_{ij}^2 \hat{u}_i^2 \xrightarrow{p} \mathbb{E}[\hat{r}_{ij}^2 u_i^2] = \mathbb{E}[\hat{r}_{ij}^2 \sigma_i^2]$$

The robust estimator substitutes $\hat{u}_i^2$ for σ_i^2 term-by-term. By the Law of Large Numbers, the average converges to the correct quantity. No assumption about the *form* of σ_i^2 is required.

Robust t and F Statistics

Once we have robust SEs, all inference proceeds with the usual formulas:

Robust t -statistic:

$$t_{\text{rob}} = \frac{\hat{\beta}_j - \beta_j^0}{\widehat{\text{se}}_{\text{rob}}(\hat{\beta}_j)} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{under } H_0 \text{ as } n \rightarrow \infty$$

Use same critical values as standard t -test.

Robust F -statistic (Wald form):

For testing q restrictions $H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$:

$$F_{\text{rob}} = \frac{1}{q} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' (\mathbf{R} \hat{\mathbf{V}}_{\text{rob}} \mathbf{R}')^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \xrightarrow{d} F_{q, \infty} \text{ under } H_0$$

Example 8.1: OLS vs. Robust SEs in Log Wage Equation

Dataset: WAGE1 ($n = 526$). Compare OLS and robust SEs (in brackets).

Log Wage Regression

Variable	Coef.	OLS SE	[Robust SE]
<i>marrmale</i>	0.213	(0.055)	[0.057]
<i>marrfem</i>	-0.198	(0.058)	[0.058]
<i>singfem</i>	-0.110	(0.056)	[0.057]
<i>educ</i>	0.0789	(0.0067)	[0.0074]
<i>exper</i>	0.0268	(0.0052)	[0.0051]
<i>exper</i> ²	-0.00054	(0.00011)	[0.00011]
<i>tenure</i>	0.0291	(0.0068)	[0.0069]
<i>tenure</i> ²	-0.00053	(0.00023)	[0.00024]
intercept	0.321	(0.100)	[0.109]
$R^2 = 0.461$			

- ▶ Robust SE for *educ*: $0.0067 \rightarrow 0.0074$ (slightly larger — mild heteroskedasticity).
- ▶ Most SEs are nearly identical — significance conclusions unchanged.
- ▶ Lesson: always report robust SEs and check if they alter inference.

Example 8.2: Robust F -Test for Race Dummies

Dataset: GPA3 ($n = 366$). Dep. var.: *cumgpa*.

Cumulative GPA Regression

Variable	Coef.	OLS SE	[Robust SE]
<i>sat</i>	0.00114	(0.00018)	[0.00019]
<i>hsperc</i>	-0.00857	(0.00124)	[0.00140]
<i>tothrs</i>	0.00250	(0.00073)	[0.00073]
<i>female</i>	0.303	(0.059)	[0.059]
<i>black</i>	-0.128	(0.147)	[0.118]
<i>white</i>	-0.059	(0.141)	[0.110]
intercept	1.47	(0.23)	[0.22]
$R^2 = 0.4006$			

Testing $H_0 : \beta_{black} = \beta_{white} = 0$: Usual $F = 0.40$ ($p = 0.672$); Robust $F \approx 0.75$ — both fail to reject. Robust SEs for race dummies are *smaller*.

Heteroskedasticity-Robust LM Test (§8-2a)

LM tests can be made robust without matrix algebra.

Robust LM Test Procedure (for testing q exclusion restrictions):

1. Estimate the **restricted** model; save residuals $\tilde{u}_i$.
2. Regress each **excluded** variable on all **included** variables; save residuals $\tilde{r}_{i\ell}$ for $\ell = 1, \dots, q$.
3. Regress the constant **1** on $\tilde{r}_{i1}\tilde{u}_i, \dots, \tilde{r}_{iq}\tilde{u}_i$ (no intercept); get SSR_0 .
4. Robust LM = $n - SSR_0 \xrightarrow{d} \chi^2_q$ under H_0 .

Example 8.3: Arrests (CRIME1)

Test $H_0 : \beta_{avg\text{sen}} = \beta_{avg\text{sen}^2} = 0$ in the arrest-rate equation.

- ▶ Usual LM = 3.52 ($p \approx 0.17$) — fail to reject
- ▶ Robust LM = 5.10 ($p \approx 0.078$) — borderline rejection at 10%
- ▶ Heteroskedasticity *distorts* the usual LM test here.

Testing for Heteroskedasticity (§8-3)

Should We Always Use Robust SEs?

Case for always-robust:

- ▶ Valid under both homo- and heteroskedasticity
- ▶ No cost to unbiasedness
- ▶ Standard in modern applied work (cross-section data)

Case for testing first:

- ▶ Robust SEs can be imprecise in small samples
- ▶ Homoskedasticity $\Rightarrow$ OLS t -test is exact in finite samples (MLR.1–6)
- ▶ Knowing the variance form enables efficient WLS

Wooldridge's recommendation:

1. **Always** report robust SEs in empirical work with cross-sectional data.
2. Run a formal test if you want to understand the *form* of

The Breusch-Pagan Test: Setup

Null hypothesis: $H_0 : \text{Var}(u \mid x_1, \dots, x_k) = \sigma^2$ (homoskedasticity)

Idea: Under H_0 , u_j^2 should be unrelated to $x_1, \dots, x_k$. Test this with a regression.

BP Test Auxiliary Regression (eq. 8.12)

$$\hat{u}_i^2 = \delta_0 + \delta_1 \mathbf{x}_{i1} + \delta_2 \mathbf{x}_{i2} + \cdots + \delta_k \mathbf{x}_{ik} + \mathbf{v}_i$$

Joint null: $H_0 : \delta_1 = \delta_2 = \dots = \delta_k = 0$ (eq. 8.13)

Why squared residuals? $\hat{u}_i^2$ is the sample proxy for $\sigma_i^2 = \text{Var}(u_i | x_i)$. If the latter depends on x , so should $\hat{u}_i^2$.

The Breusch-Pagan Test: Test Statistics

Two equivalent test statistics:

F-statistic from the auxiliary regression [8.12]:

$$F = \frac{R_{\hat{u}^2}^2/k}{(1 - R_{\hat{u}^2}^2)/(n - k - 1)} \sim F_{k, n-k-1} \quad \text{under } H_0$$

LM statistic (Lagrange Multiplier):

$$LM = n \cdot R_{\hat{u}^2}^2 \xrightarrow{d} \chi_k^2 \quad \text{under } H_0 \quad (\text{eq. 8.16})$$

where $R_{\hat{u}^2}^2$ is the R^2 from regressing $\hat{u}_i^2$ on $x_1, \dots, x_k$.

Limitation: BP detects only *linear* dependence of σ^2 on x .

Example 8.4: BP Test — Levels vs. Logs

Level Model (HPRICE1)

$$\begin{aligned}\widehat{price} &= -21.77 \\ &+ 0.00207 \text{ lotsize} \\ &+ 0.123 \text{ sqrft} + 13.85 \text{ bdrms} \\ n &= 88, R^2 = 0.672\end{aligned}$$

BP $F = 5.34$, $p = 0.002$

BP $LM = 14.09$, $p = 0.003$

Reject H_0 — heteroskedastic.

Log-Log Model (HPRICE1)

$$\begin{aligned}\log(\widehat{price}) &= -1.30 \\ &+ 0.168 \log(\text{lotsize}) \\ &+ 0.700 \log(\text{sqrft}) + 0.037 \text{ bdrms} \\ n &= 88, R^2 = 0.643\end{aligned}$$

BP $F = 1.41$, $p = 0.245$

BP $LM = 4.22$, $p = 0.239$

Fail to reject H_0 .

Log-transforming the dependent variable *stabilises the variance*. Level models for prices, wages, and income frequently exhibit heteroskedasticity that disappears in log form.

The White Test: Motivation

Problem with BP: Misses heteroskedasticity that depends on *squares and cross-products* of regressors.

White (1980) proposed a more general auxiliary regression:

White Test — Full Form (eq. 8.19)

With regressors $x_1, x_2, \dots, x_k$, regress $\hat{u}_i^2$ on all unique terms:

$$x_1, x_2, \dots, x_k, \quad x_1^2, x_2^2, \dots, x_k^2, \quad x_1x_2, x_1x_3, \dots, x_{k-1}x_k$$

Joint F -test or $LM = nR_{\hat{u}^2}^2 \xrightarrow{d} \chi_m^2$ where m = number of slope terms.

Cost of the full White test: With k regressors, there are $k + k + \binom{k}{2} = k(k+3)/2$ terms. For $k = 5$, that is 20 regressors in the auxiliary regression —

The White Test: Parsimonious Special Case

Insight: The fitted value $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}$ is a linear combination of all x_j , so $\hat{y}_i^2$ is a linear combination of all cross-products.

White Test — Special Case (eq. 8.20)

Regress $\hat{u}_i^2$ on the OLS fitted values and their squares:

$$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + \text{error}$$

$H_0 : \delta_1 = \delta_2 = 0.$ $LM = nR_{\hat{u}^2}^2 \xrightarrow{d} \chi_2^2$ — only **2 degrees of freedom** regardless of k .

Advantage: Degrees of freedom do not grow with k . Powerful for detecting common heteroskedasticity patterns (variance increasing with $\hat{y}$).

Example 8.5: White Test on the Log Housing Price Equation

Recall: BP test on the log-log model failed to reject homoskedasticity ($p = 0.239$).

Special White Test

Regress $\hat{u}_i^2$ on $\widehat{\log(\text{price})}_i$ and $\widehat{\log(\text{price})}_i^2$:

$$R_{\hat{u}^2}^2 = 0.0392$$

$$LM = 88 \times 0.0392 = 3.45, \quad p\text{-value} = 0.178 \quad (2 \text{ d.f.})$$

- ▶ Fails to reject at 10% level (slightly more evidence than BP, but inconclusive).
- ▶ Both tests agree: the log-log model is approximately homoskedastic.

Weighted Least Squares (§8-4)

The Logic of WLS: Re-weighting Observations

Motivation: If we know how variance varies, we can do better than OLS.

WLS Setup (eq. 8.21)

Suppose the heteroskedasticity has the form:

$$\text{Var}(u_i \mid \mathbf{x}_i) = \sigma^2 h_i, \quad h_i = h(x_{i1}, \dots, x_{ik}) > 0$$

where h_i is known (or estimable).

WLS minimises the weighted sum of squared residuals (eq. 8.27):

$$\sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_k x_{ik})^2}{h_i}$$

WLS as OLS on a Transformed Model

Divide the regression equation through by $\sqrt{h_i}$:

Starting from (eq. 8.24):

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

Dividing by $\sqrt{h_i}$ gives the **transformed model** (eq. 8.25–8.26):

$$\underbrace{\frac{y_i}{\sqrt{h_i}}}_{y_i^*} = \beta_0 \underbrace{\frac{1}{\sqrt{h_i}}}_{x_{i0}^*} + \beta_1 \underbrace{\frac{x_{i1}}{\sqrt{h_i}}}_{x_{i1}^*} + \cdots + \underbrace{\frac{u_i}{\sqrt{h_i}}}_{u_i^*}$$

The transformed error satisfies:

$$\text{Var}(u_i^* \mid x_i) = \frac{\text{Var}(u_i \mid x_i)}{h_i} = \frac{\sigma^2 h_i}{h_i} = \sigma^2 \quad (\text{homoskedastic!})$$

WLS: Key Properties and Warning

If h_i is correctly specified:

- ▶ WLS is **unbiased** (linear transformation of the data)
- ▶ WLS is **BLUE** — lower variance than OLS
- ▶ WLS standard errors (from the transformed regression) are valid

Warning — misspecification risk:

If h_i is *wrong*, WLS is still consistent but:

- ▶ WLS standard errors may be **invalid**
- ▶ WLS may even be **less efficient** than OLS

Safe practice: After WLS, always compute *heteroskedasticity-robust SEs* for the WLS estimates. These are valid even if h_i is misspecified.

Example 8.6: Financial Wealth with WLS

Setup: Assume $\text{Var}(u_i \mid \text{inc}_i) = \sigma^2 \cdot \text{inc}_i$, so $h_i = \text{inc}_i$.

Financial Wealth Equations (401KSUBS)

Variable	OLS	WLS ($w_i = 1/\text{inc}_i$)
<i>inc</i>	0.821 (0.104)	0.787 (0.063)
intercept	-10.57 (2.53)	-9.58 (1.65)
<i>inc</i> (expanded)	0.771 (0.100)	0.740 (0.064)
$(\text{age} - 25)^2$	0.0251 (0.0043)	0.0175 (0.0019)
<i>male</i>	2.48 (2.06)	1.84 (1.56)
<i>e401k</i>	6.89 (2.29)	5.19 (1.70)

- ▶ WLS standard errors are consistently **smaller** — efficiency gain from correct weighting.
- ▶ 401(k) eligibility coefficient drops from 6.89 (OLS) to 5.19 (WLS) — *economically meaningful*.
- ▶ WLS is preferred here assuming $\text{Var}(u \mid \text{inc}) \propto \text{inc}$.

Feasible GLS: When h_i Must Be Estimated (§8-4b)

In practice, the variance function $h(x)$ is almost never known exactly.

FGLS: Exponential Variance Function (eq. 8.30)

Assume:

$$\text{Var}(u_i | \mathbf{x}_i) = \sigma^2 \exp(\delta_0 + \delta_1 \mathbf{x}_{i1} + \cdots + \delta_k \mathbf{x}_{ik})$$

Taking logs of $u_i^2 \approx \sigma^2 \exp(\delta_0 + \delta_1 \mathbf{x}_{i1} + \cdots + \delta_k \mathbf{x}_{ik}) \cdot \epsilon_i$:

$$\log(\hat{u}_i^2) = \alpha_0 + \delta_1 \mathbf{x}_{i1} + \cdots + \delta_k \mathbf{x}_{ik} + \text{error} \quad (\text{eq. 8.32})$$

Estimated weights: $\hat{h}_i = \exp(\hat{g}_i)$ where $\hat{g}_i$ are fitted values from [8.32].

Feasible GLS: Step-by-Step Algorithm

FGLS Procedure:

1. Estimate main equation by **OLS**; obtain residuals $\hat{u}_i$.
2. Compute $\log(\hat{u}_i^2)$.
3. Regress $\log(\hat{u}_i^2)$ on $x_{i1}, \dots, x_{ik}$; save fitted values $\hat{g}_i$.
4. Form estimated weights: $\hat{h}_i = \exp(\hat{g}_i)$.
5. Apply **WLS** with weights $w_i = 1/\hat{h}_i$.
6. Report **robust SEs** for the WLS step (in case h_i is misspecified).

Properties of FGLS:

- **Consistent and asymptotically more efficient** than OLS (if form is correct)

Example 8.7: Cigarette Demand — OLS vs. FGLS

Dataset: SMOKE ($n = 807$). BP test: $LM = 32.3$ ($p \approx 0$) — strong heteroskedasticity.

Cigarette Demand: OLS vs. FGLS

Variable	OLS	FGLS
$\log(\text{income})$	0.880 (0.728)	1.30 (0.44)
$\log(\text{cigprice})$	-0.751 (5.773)	-2.94 (4.46)
educ	-0.501 (0.167)	-0.463 (0.120)
age	0.771 (0.160)	0.482 (0.097)
age^2	-0.0090 (0.0017)	-0.0056 (0.0009)
restaurn	-2.83 (1.11)	-3.46 (0.80)
R^2	0.053	0.113

- ▶ FGLS standard errors are **uniformly smaller** — efficiency gain from re-weighting.
- ▶ Income elasticity doubles from 0.88 to 1.30 (more precisely estimated under FGLS).
- ▶ Restaurant restrictions: -3.46 cigarettes/day (larger and more precisely estimated).

Estimation Strategy Summary

Deciding which estimator and standard errors to use:

Situation	Estimator	Standard Errors
Homoskedasticity confirmed	OLS	Usual OLS SEs
Heterosk., form unknown	OLS	White robust SEs
Heterosk., form known	WLS	Usual WLS SEs
Heterosk., form estimated	FGLS	Robust WLS SEs

When in doubt: **OLS + robust SEs** is the defensible baseline. Pursue WLS/FGLS only when you have a strong theoretical or empirical reason to believe a specific variance function.

The Linear Probability Model Revisited (§8-5)

LPM Heteroskedasticity: The Theoretical Result

Recall from Lecture 6: the LPM has *built-in* heteroskedasticity.

Proof: For the LPM with $y_i \in \{0, 1\}$, given x_i , y_i is Bernoulli with mean $p_i = \mathbb{E}[y_i \mid x_i]$.

Therefore: $\text{Var}(y_i \mid x_i) = p_i(1 - p_i) = p(x_i)[1 - p(x_i)]$ (eq. 8.45)

Since $p(x_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}$ depends on x_i , so does the variance. MLR.5 is **always violated** in the LPM.

Natural WLS weights for LPM (eq. 8.47):

$$\hat{h}_i = \hat{y}_i(1 - \hat{y}_i)$$

Constraint: WLS is only feasible if all $\hat{y}_i \in (0, 1)$. If any $\hat{y}_i \leq 0$ or $\hat{y}_i \geq 1$, weights are non-positive and WLS fails.

Example 8.8: Married Women's Labor Force Participation

Dataset: MROZ ($n = 753$). $y = inlf$ (1 = in labor force).

LPM for Labor Force Participation

$$\begin{aligned}\widehat{inlf} = & 0.586 - 0.0034 \textit{nwifeinc} + 0.038 \textit{educ} + 0.039 \textit{exper} \\ & (0.154) \quad (0.0014) \quad (0.007) \quad (0.006) \\ & [0.151] \quad [0.0015] \quad [0.007] \quad [0.006] \\ & -0.00060 \textit{exper}^2 - 0.016 \textit{age} - 0.262 \textit{kidslt6} + 0.013 \textit{kidsge6} \\ & n = 753, \quad R^2 = 0.264\end{aligned}$$

OLS SEs in (); robust SEs in []

- ▶ OLS and robust SEs are very similar — heteroskedasticity is mild here.
- ▶ Each young child (*kidslt6*) reduces LFP probability by **26.2 pp** (large!).
- ▶ Each year of education raises LFP by **3.8 pp**.
- ▶ Conclusion: OLS + robust SEs is sufficient; WLS offers little gain.

Example 8.9: PC Ownership — OLS vs. WLS for LPM

Dataset: GPA1 ($n = 141$). $y = PC$ (1 = owns computer).

PC Ownership: OLS vs. WLS

Variable	OLS	WLS ($w_i = 1/\hat{h}_i$)
<i>hsGPA</i>	0.065 (0.137)	0.033 (0.130)
<i>ACT</i>	0.00065 (0.0155)	0.0043 (0.0155)
<i>parcoll</i>	0.221 (0.093)	0.215 (0.086)
intercept	-0.0004 (0.490)	0.026 (0.477)
R^2	0.0415	0.0464

- ▶ OLS and WLS give very similar results — no efficiency concern here.
- ▶ *parcoll* is the only significant predictor: parental college degree raises PC ownership by ≈ 22 pp.
- ▶ With mild heteroskedasticity and small n , OLS + robust SEs is preferred.

Summary and Decision Framework

Three-Step Decision Framework

Step 1 — Always estimate by OLS first.

Report OLS estimates with **heteroskedasticity-robust SEs**.

OLS is unbiased and consistent under MLR.1–4. Robust SEs ensure valid inference.

Step 2 — Test for heteroskedasticity (optional but informative).

Run **BP test** (χ_k^2) and/or **White test** (χ_2^2 in special case).

If rejected: proceed to Step 3. If not: OLS + usual SEs suffices (but robust SEs still safe).

Step 3 — Pursue efficiency if heteroskedasticity is confirmed.

- ▶ Known h_i : use WLS with usual WLS SEs.

Key Concepts Reviewed

The Problem: Heteroskedasticity ($\sigma_i^2 \neq \sigma^2$) does not bias $\hat{\beta}_j$ but invalidates standard errors and test statistics.

The Fix for Inference: Robust (White) SEs — valid asymptotically under any heteroskedasticity. Formula: $\widehat{\text{Var}}(\hat{\beta}_j) = \sum_i \hat{r}_{ij}^2 \hat{u}_i^2 / \text{SSR}_j^2$.

Testing: BP ($\hat{u}^2$ on x_j, χ_k^2) or White ($\hat{u}^2$ on $\hat{y}, \hat{y}^2, \chi_2^2$).

The Fix for Efficiency: WLS/FGLS — achieves BLUE if h_i correctly specified. Requires variance function $h_i = \text{Var}(u_i | x_i) / \sigma^2$ known or estimated.

Bridge to Lecture 8: Specification and Data Issues

Lecture 7 (today) — violation of MLR.5 only:

- ▶ OLS is still unbiased and consistent
- ▶ Only inference (SEs, tests) is affected
- ▶ Fix: robust SEs or WLS

Lecture 8 (Chapter 9) — violation of MLR.4 ($\mathbb{E}[u | x] \neq 0$):

- ▶ **Omitted variable bias** — $\hat{\beta}_j$ is biased and inconsistent
- ▶ **Functional form misspecification** — RESET test
- ▶ **Measurement error** — attenuation bias
- ▶ **Missing data and selection** — selection bias